

Year 10 -Pathway 2
Mini Test 5.1 -Semester 2
Statistics

Resources allowed: Calculator and 1 x A4 page one side

Name: ANSWERS Class: _____ Time: 30 minutes Total Marks: ____/16

Show working and answers on this sheet. Show working in sufficient detail to support your answers.
Incorrect answers without supporting reasoning may be allocated zero marks.

Question 1

(4 Marks)

Complete the calculations for the following measures of central tendency for the data set below:

1, 5, 5, 7, 8, 10

Mean: $1 + 5 + 5 + 7 + 8 + 10$
 $= 36 \div 6$
 $= 6$

Median: $\frac{5 + 7}{2}$
 $= 6$

Mode: 5

Range: $10 - 1$
 $= 9$

Question 2

(4 Marks)

Michael drinks the following number of cups of coffee each day in a week:

4, 5, 3, 6, 4, 3, 3

a) How many cups of coffee does he drink in a week?

$$4 + 5 + 3 + 6 + 4 + 3 + 3 = 28$$

b) What is the mean number of cups he has in a week?

$$28 \div 7 = 4$$

c) What is the mode for this data set? 3

d) What is the median number of cups of coffee Michael drinks in a week?

3, 3, 3, 4, 4, 5, 6

4

Question 3**(3 Marks)**

The results of a mathematics test are represented by this stem-and-leaf plot:

Key: 1|6 = 16

Stem	Leaf
0	4 9
1	0 3 7 8
2	1 1 1 3 4 5
3	2 7
4	1 3 3 5 7

a) Find the mean, median and mode.

4, 9, 10, 13, 17, 18, 21, 21, 21, 23, 24, 25, 32, 37, 41, 43, 43, 43, 47

$$\text{mean} = 494 \div 19$$

$$= 26$$

$$\text{median} = 23$$

$$\text{mode} = 21$$

Question 4**(5 Marks)**

Sally is designing a website to advertise her services as a stuntwoman. She wants to be paid as much as possible and so is giving her average charge based on the last 8 films she has done. Calculate the mean, mode, median and range and decide which would be the best charge to put on her website.

Earning per film: \$1000, \$8000, \$7500, \$3200, \$5000, \$8000, \$600, \$200

$$\text{mean} = 33500 \div 8$$

$$= \$4187.50$$

$$\text{median} = 200, 600, 1000, 3200, 5000, 7500, 8000, 8000$$

$$= \frac{3200 + 5000}{2}$$

$$= \$4100$$

$$\text{mode} = \$8000$$

$$\text{range} = 8000 - 200$$

$$= \$7800$$

Mode would be the measure she would use, as it is the highest charge out of all of the measures.

~ End of Test ~